Lab 4
Basics of Signals:
Frequency

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I. Introduction

This lab had two main objectives. The first was to get used to working with time-varying signals using the function generator and oscilloscope, looking at sinusoidal, square, and ramp waveforms. They were viewed both on the screen and by ear through a speaker, which also tested the limits of human hearing. The second objective was to use LTspice to perform transient and Fourier (FFT) analysis on circuits driven by a pulse source and by sinusoidal sources, to see how a signal's shape relates to its frequency components.

II. Results

1. Frequency

A sinusoidal signal at 600 Hz with a 0.10 V peak-to-peak amplitude was set on the function generator and observed on the oscilloscope.

From the oscilloscope measurements, the measured frequency was also very close to 600 Hz, in agreement with the setting on the function generator.

Changing the signal's amplitude and frequency also affected the sound coming out of the speakers. As the amplitude increased, the sound got louder and as it decreased, it got softer. As the frequency increased, the sound pitch got higher and as it decreased, the pitch got lower.

The lowest frequency we could hear from the speaker was 20 Hz. My partner and I differed slightly in our top limit of hearing frequency although it was comparable. My partner was able to hear 17 kHz while I heard 15 kHz. This falls within the range of normal human hearing of 20 Hz to 20 kHz. Generally, the sound's loudness increases with frequency, but we observed it was the loudest around the 1-3 kHz range. This can be explained due to human hearing being most

sensitive around the 2-5 kHz range. Our results demonstrated deviations between individuals' hearing but could also be due to equipment.

We worked with another group in an attempt to match our speaker's frequencies through pure sound. We set the sinusoidal wave frequency to 750 Hz, and the other group tried to match this. They arrived at 763 Hz, which was very similar to ours.

The three different waves: sinusoidal, square, and ramp, were compared with the frequency set to 600 Hz. We ordered them in terms of their pitch as that was what was apparent to us. The square waves had the highest pitch, followed by ramp waves, and ending with sinusoidal waves. This is due to square and ramp waves having higher frequency harmonics.

We repeated the process but now at 6 kHz. Ramp waves overtook square waves for having the higher pitch with sinusoidal waves still being last. This is because ramp waves' harmonic levels sit at higher frequencies, and square wave harmonics can fall out of audible range.

2. LTspice, transient and Fourier analysis

The circuit in Fig. 1 was built in LTspice using a PULSE source.

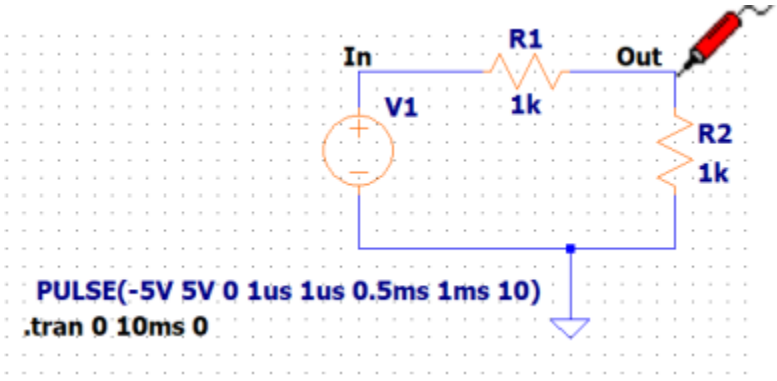


Figure 1: Circuit diagram with pulse source [Ref. 1]

A transient analysis was run and the voltage across R2 was plotted as a function of time, shown in Fig. 2 alongside the voltage across R1. The waveform is a symmetric square wave alternating between −5 V and 5 V. The FFT of the output, shown in Fig. 3, was then taken with the frequency axis set from 0 to 30 kHz in 1 kHz steps and the magnitude axis set from −50 dB to 20 dB in 10 dB steps. The spectrum shows a strong component at the fundamental frequency of 1 kHz, with decreasing peaks also at higher harmonics. The odd harmonics stand out more than the even ones, consistent with the Fourier series of a square wave.

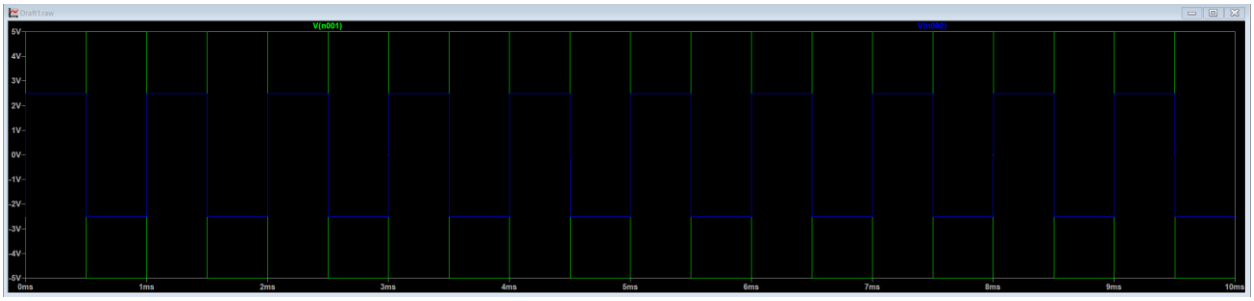


Figure 2: V vs. Time for circuit in Fig. 1

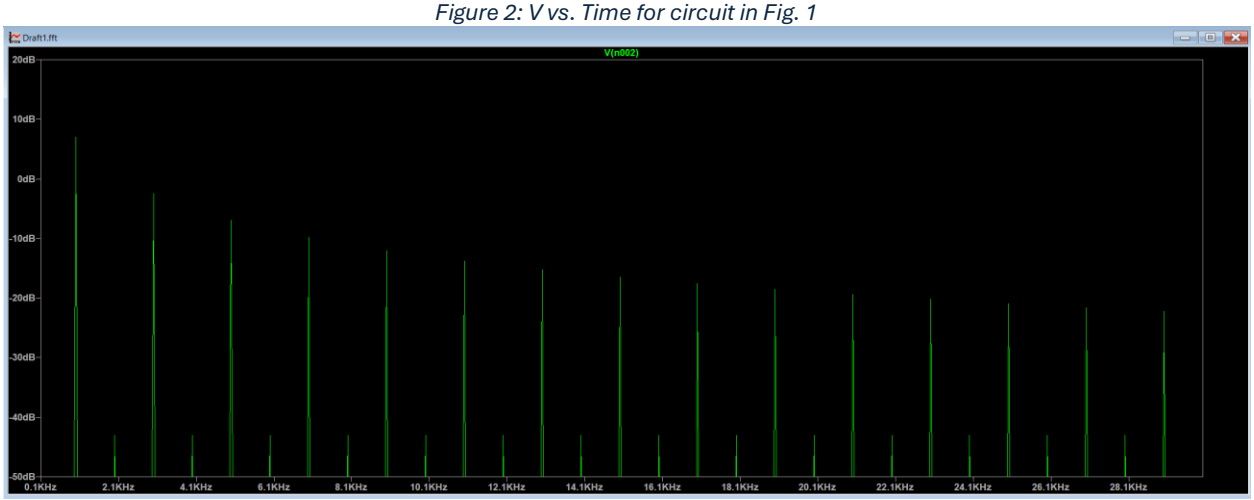


Figure 3: FFT of output voltage for Fig. 1

The pulse width (Ton) was shortened to 0.05 ms, with the same axis settings on the FFT with the results shown in Fig. 4. More high frequency components appeared, and narrow pulses added more harmonics.

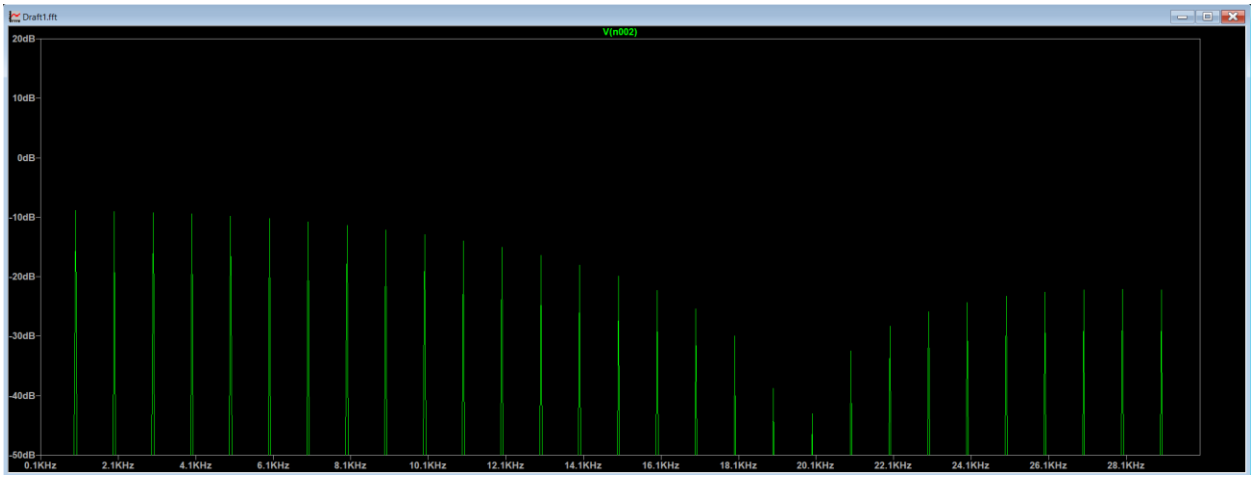


Figure 4: FFT of Fig. 3 with pulse width shortened to Ton = 0.05ms

The source was then changed to a 1000 Hz sinusoid with a 5 V amplitude. The resulting FFT, shown in Fig. 5, has only a single component at 1 kHz, as expected since a pure sinusoid has no harmonic content. It is already a single frequency.



Figure 5: FFT of output voltage for 1 kHz, 5V sinusoidal wave

Taken together, Figs. 3-5 show that any periodic signal that is not a pure sinusoid is built from a fundamental frequency plus a series of harmonics. A sinusoid is one periodic waveform made of a single frequency with no additional harmonics.

Next, the circuit in Fig. 6 was built, with three sinusoidal voltage sources V1, V2, and V3 at fundamental frequencies of 1 kHz, 3 kHz, and 5 kHz. The output node Vo is the average of the three source voltages.

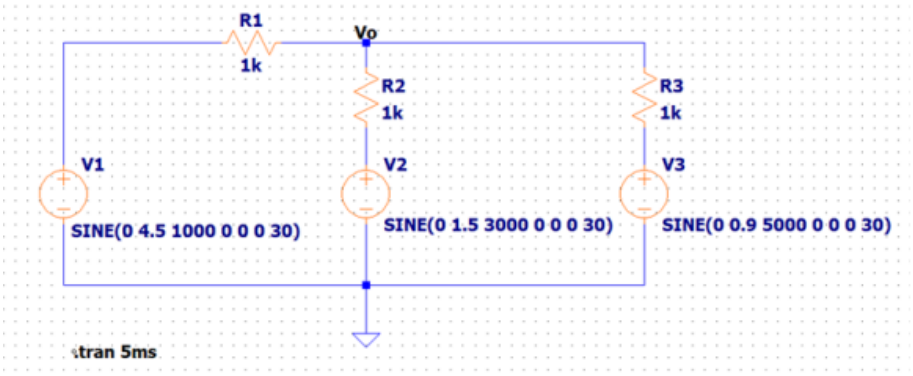


Figure 3. Circuit for part 2.6

Figure 6: Circuit diagram at fundamental frequencies: 1 kHz, 3 kHz, 5 kHz [Ref. 1]

The resulting Vo(t), shown in Fig. 7, resembles a rounded version of a square wave rather than a smooth sinusoid, with small ripples on the flat sections. The FFT of Vo, shown in Fig. 8, has three peaks at 1 kHz, 3 kHz, and 5 kHz, confirming that the frequency content of Vo is exactly the three source frequencies that were added together.

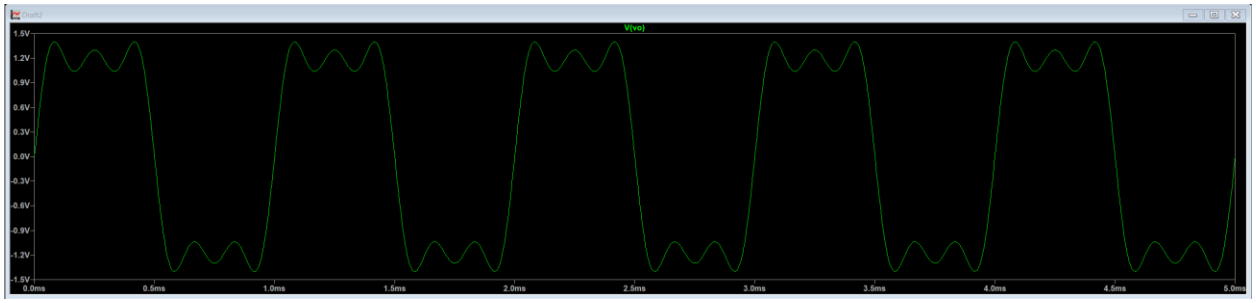


Figure 7: Vo(t) for circuit in Fig. 6

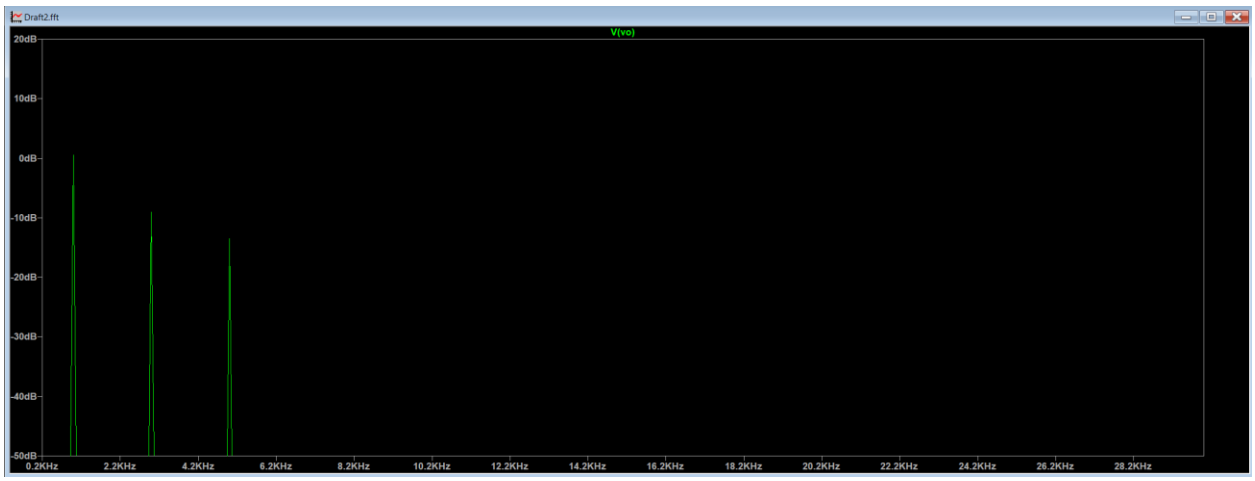


Figure 8: FFT of Vo for circuit in Fig. 6

The amplitudes and phases of V1, V2, and V3 were then changed as shown in Fig. 9: V2's amplitude was changed to 0.5 V with a 180° phase, and V3's amplitude was reduced to 0.18 V.

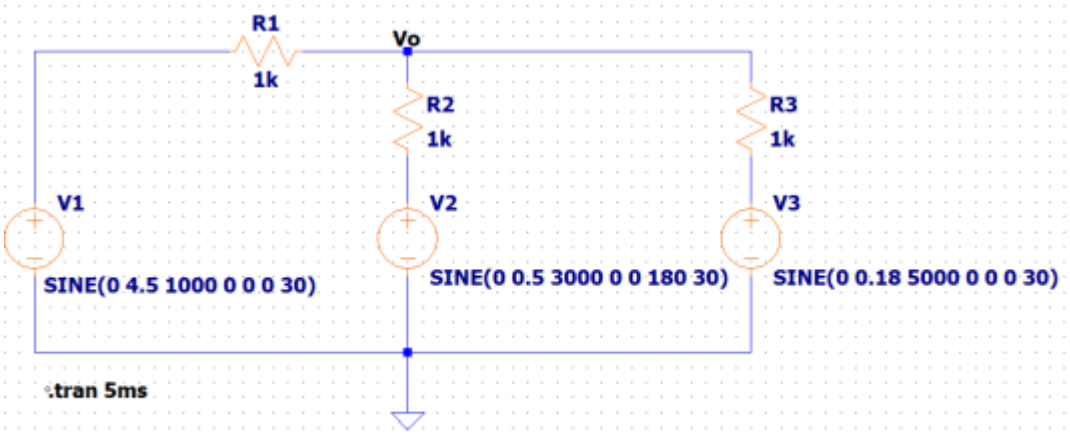


Figure 4. Circuit for part 2.7

Figure 9: Circuit for part 2.7 [Ref. 1] with modified sinusoidal sources

The new Vo(t), shown in Fig. 10, is a clean, smooth triangle wave. Comparing Figs. 7 and 10 shows that it is not simply which frequencies are present that determines a periodic waveform's shape, but the specific amplitude and phase of each harmonic relative to the others. The FFT in

Fig. 11 again confirms peaks at 1 kHz, 3 kHz, and 5 kHz, now with the smaller relative weight of the 3 kHz and 5 kHz components visible.

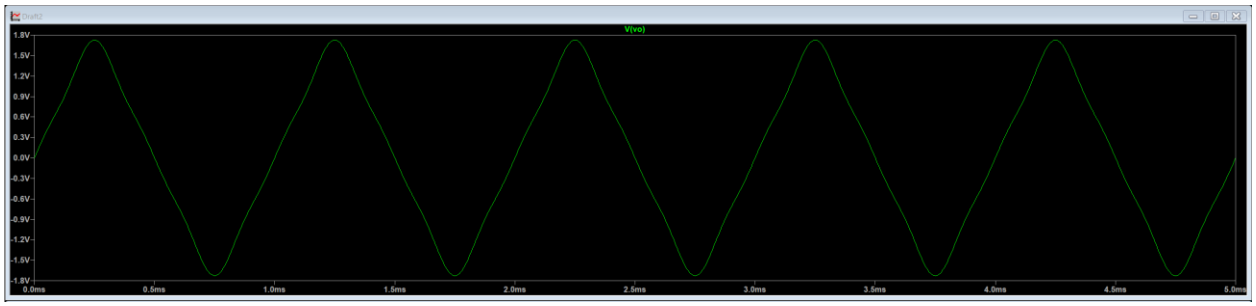


Figure 10: Vo(t) for circuit in Fig. 9



Figure 11: FFT of Vo for Fig. 9

Finally, R2 in the circuit from part 2.1 (Fig. 1) was replaced with a 1 µF capacitor and the square-wave analysis from part 2.2 was repeated. Fig. 12 shows instead of following the sharp square-wave transitions, the capacitor voltage rises and falls in a near-triangular ramp, since the capacitor cannot charge or discharge instantaneously. The FFT of the capacitor voltage, shown in Fig. 13, decays much faster with frequency than the resistor-only case in Fig. 3.

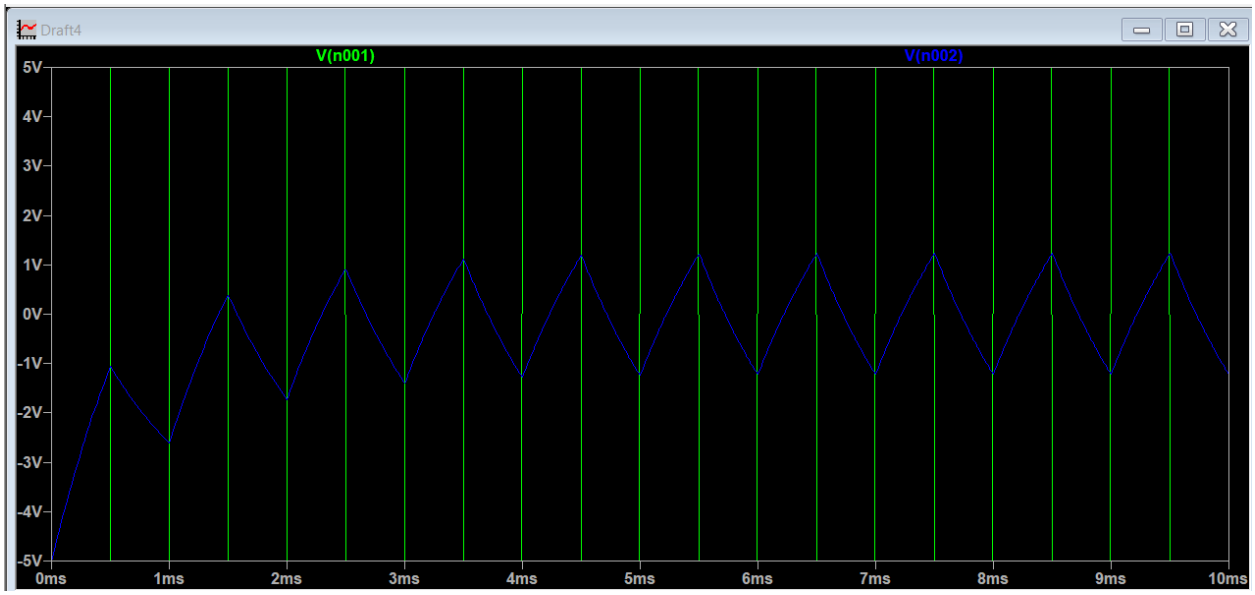


Figure 12: V vs time for circuit in Fig. 1 but with R2 changed to 1uF capacitor

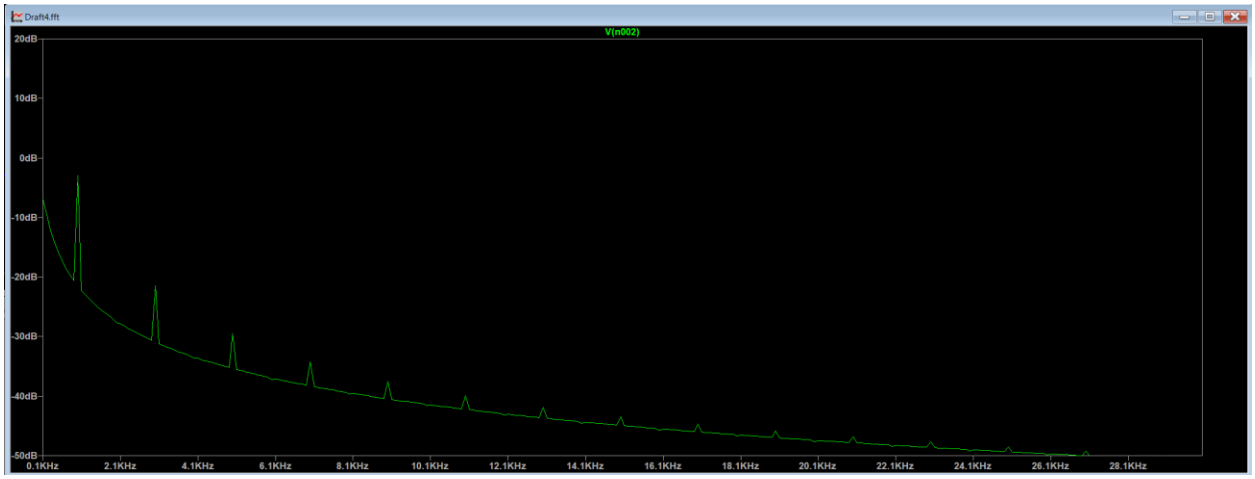


Figure 13: FFT of capacitor voltage from Fig. 12

III. Discussions and Conclusions

In this lab we explored how signals look and sound in both the time and frequency domain. On the speaker, the sine wave sounded pure while the square and ramp waves sounded harsher due to their harmonics. My partner and I had slightly different hearing limits, and the frequency-matching exercise with the other group came close. The ordering of pitch between the three

waveforms changed between 600 Hz and 6 kHz because the square wave's harmonics moved out of hearing range sooner than the ramp waves did.

In LTspice, the Fourier analysis backed this up. The square wave's FFT showed a fundamental plus odd harmonics, while the sine wave had only one frequency component. Narrowing the pulse width added more high-frequency harmonics. Combining a fundamental with its harmonics could produce either a square-like or a triangle-like waveform depending on the amplitude and phase of each harmonic, even when the same three frequencies were used. Replacing the resistor with a capacitor filtered out the higher harmonics, confirming its role as a low-pass filter.

Overall, the lab connected what was heard, seen on the oscilloscope, and shown in the FFT, all pointing to the same idea: periodic signals are built from a fundamental frequency and its harmonics, and these tools for moving between the time and frequency domains will be useful throughout the rest of the course.

IV. References

[1] Dr. Iman Salama. “Lab 4 – Basics of Signals: Frequency.” Northeastern University. 19 September 2023.